

PRODUCT AND CATEGORY MANAGEMENT

Project Title: Vilvah

1. Introduction

The Vilvah Natural Skincare and Personal Care Database Management System is designed to manage product and category information efficiently. The system uses related tables to store and organize product and category details. Primary key and foreign key constraints are used to maintain data integrity and establish relationships between the entities.

SQL operations such as **INSERT**, **UPDATE**, and **DELETE** are performed to manage product information. Category-wise reports can be generated using SQL to analyze products, prices, and stock information.

2. Objectives

- Design tables for Product and Category entities.
- Define primary key and foreign key relationships.
- Store product name, category, price, and stock details.
- Perform product insertion, updating, and deletion.
- Generate category-wise product reports.
- Maintain data integrity using constraints.

3. Relationship

The **Vilvah database** contains relationships between customers, products, categories, suppliers, orders, order details, payments and admins.

CUSTOMER (1) ————— (M) ORDER

SUPPLIER (1) ————— (M) PRODUCT

CATEGORY (1) ————— (M) PRODUCT

PRODUCT (1) ————— (M) ORDER DETAILS

ORDER (1) ————— (M) ORDER DETAILS

ORDER (1) ————— (1) PAYMENT

PRODUCT (1) ————— (1) INVENTORY

ADMIN (1) ————— (M) PRODUCT

ADMIN (1) ————— (M) ORDER

4. Table Design

4.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER(10)	Primary Key
Category_Name	VARCHAR2(50)	NOT NULL, UNIQUE
Category_Description	VARCHAR2(200)	—
Status	VARCHAR2(20)	CHECK

4.2 Product Table

Attribute	Data Type	Constraint
Product_ID	NUMBER(10)	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Category_ID	NUMBER(10)	Foreign Key
Supplier_ID	NUMBER(10)	Foreign Key
Price	NUMBER(10,2)	NOT NULL
Weight	VARCHAR2(20)	—
Ingredients	VARCHAR2(500)	—
Skin_Type	VARCHAR2(30)	—
Brand	VARCHAR2(50)	—
Expiry_Date	DATE	—

5. Table Creation

5.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER(10),  
3     Category_Name VARCHAR2(50) NOT NULL,  
4     Category_Description VARCHAR2(200),  
5     Status VARCHAR2(20),  
6     CONSTRAINT PK_CATEGORY PRIMARY KEY (Category_ID),  
7     CONSTRAINT UQ_CATEGORY_NAME UNIQUE (Category_Name),  
8     CONSTRAINT CHK_CATEGORY_STATUS  
9     CHECK (Status IN ('Active', 'Inactive'))  
10 );
```

Table created.

5.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID NUMBER(10),  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Category_ID NUMBER(10) NOT NULL,  
5     Supplier_ID NUMBER(10),  
6     Price NUMBER(10,2) NOT NULL,  
7     Weight VARCHAR2(20),  
8     Ingredients VARCHAR2(500),  
9     Skin_Type VARCHAR2(30),  
10    Brand VARCHAR2(50),  
11    Expiry_Date DATE,  
12    CONSTRAINT PK_PRODUCT PRIMARY KEY (Product_ID),  
13    CONSTRAINT FK_PRODUCT_CATEGORY  
14    FOREIGN KEY (Category_ID)  
15    REFERENCES Category(Category_ID),  
16    CONSTRAINT CHK_PRODUCT_PRICE  
17    CHECK (Price > 0)  
18 );
```

Table created.

6 Data Insertion

6.1 Category Records

```
SQL> INSERT INTO Category VALUES(1, 'Face Care', 'Products for facial skincare', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(2, 'Body Care', 'Products for body skincare', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(3, 'Hair Care', 'Products for healthy hair and scalp', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(4, 'Bath and Shower', 'Products for bathing and cleansing', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(5, 'Lip Care', 'Products for lip moisturizing', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(6, 'Skin Care', 'Products for overall skin health', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(7, 'Face Masks', 'Products for face treatment', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(8, 'Moisturizers', 'Products for skin hydration', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(9, 'Cleansers', 'Products for cleansing skin', 'Active');
1 row created.

SQL> INSERT INTO Category VALUES(10, 'Sunscreen', 'Products for sun protection', 'Active');
1 row created.
```

6.2 Product Records

```
SQL> INSERT INTO Product VALUES(101, 'Neem Face Wash', 1, 1, 299.00, '100ml', 'Neem, Aloe Vera, Tea Tree', 'All Skin', 'Vilvah', TO_DATE('15-06-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(102, 'Aloe Vera Body Lotion', 2, 2, 399.00, '200ml', 'Aloe Vera, Shea Butter, Coconut Oil', 'Dry Skin', 'Vilvah', TO_DATE('20-07-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(103, 'Herbal Hair Oil', 3, 3, 349.00, '100ml', 'Coconut Oil, Hibiscus, Amla', 'All Hair Types', 'Vilvah', TO_DATE('10-08-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(104, 'Natural Bath Soap', 4, 4, 199.00, '100g', 'Coconut Oil, Neem, Turmeric', 'All Skin', 'Vilvah', TO_DATE('25-05-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(105, 'Shea Butter Lip Balm', 5, 5, 149.00, '10g', 'Shea Butter, Beeswax, Vitamin E', 'All Skin', 'Vilvah', TO_DATE('18-09-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(106, 'Vitamin C Face Serum', 6, 6, 599.00, '30ml', 'Vitamin C, Hyaluronic Acid, Aloe Vera', 'Normal Skin', 'Vilvah', TO_DATE('12-10-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(107, 'Turmeric Face Mask', 7, 7, 299.00, '50g', 'Turmeric, Sandalwood, Multani Hitti', 'Oily Skin', 'Vilvah', TO_DATE('05-11-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(108, 'Coconut Moisturizer', 8, 8, 449.00, '100ml', 'Coconut Oil, Aloe Vera, Shea Butter', 'Dry Skin', 'Vilvah', TO_DATE('22-12-2028', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(109, 'Aloe Vera Cleanser', 9, 9, 329.00, '150ml', 'Aloe Vera, Green Tea, Vitamin E', 'Sensitive Skin', 'Vilvah', TO_DATE('14-01-2029', 'DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Product VALUES(110, 'Mineral Sunscreen SPF 50', 10, 10, 699.00, '50g', 'Zinc Oxide, Aloe Vera, Vitamin E', 'All Skin', 'Vilvah', TO_DATE('28-02-2029', 'DD-MM-YYYY'));
1 row created.
```

7. Data Retrieval

7.1 Display Category Records

```
SQL> SELECT * FROM Category;

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
STATUS
-----
          1 Face Care
Products for facial skincare
Active

          2 Body Care
Products for body skincare
Active

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
STATUS
-----
          3 Hair Care
Products for healthy hair and scalp
Active

          4 Bath and Shower
Products for bathing and cleansing

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
STATUS
-----
Active

          5 Lip Care
Products for lip moisturizing
Active

          6 Skin Care

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
STATUS
-----
Products for overall skin health
Active

          7 Face Masks
Products for face treatment
Active

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
STATUS
-----
          8 Moisturizers
Products for skin hydration
Active

          9 Cleansers
Products for cleansing skin
Active

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
STATUS
-----
          10 Sunscreen
Products for sun protection
Active

10 rows selected.
```

7.2 Display Product Records

SQL> SELECT * FROM Product;				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
183				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
Nour Face Wash				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
1				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
2				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
Nour, Aloe Vera, Tea Tree				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
All Skin				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
Vilvah				
15-FEB-28				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
182				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
181				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
180				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
179				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
178				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
177				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
176				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
175				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
174				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
173				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
172				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
171				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
170				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
169				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
168				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
167				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
166				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
165				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
164				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
163				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
162				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
161				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
160				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
159				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
158				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
157				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
156				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
155				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
154				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
153				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
152				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
151				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
150				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
149				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
148				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
147				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
146				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
145				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
144				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
143				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
142				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
141				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
140				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
139				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
138				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
137				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
136				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
135				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
134				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
133				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
132				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
131				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
130				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
129				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
128				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
127				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
126				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
125				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
124				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
123				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
122				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
121				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
120				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
119				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
118				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
117				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
116				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
115				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
114				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
113				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
112				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
111				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
110				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
109				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
108				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
107				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
106				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
105				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
104				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				
103				
PRODUCT_ID				
PRODUCT_NAME				
CATEGORY_ID	SUPPL_ID	PRICE	WEIGHT	
INGREDIENTS				
SKIN_TYPE				
BRAND				
EXP300V_SA				

8. Product Update

```
SQL> UPDATE Product
  2  SET Price = 379.00
  3  WHERE Product_ID = 101;

1 row updated.

SQL> UPDATE Product
  2  SET Price = 649.00
  3  WHERE Product_ID = 106;

1 row updated.
```

8.1 Product Deletion

```
SQL> DELETE FROM Product
  2  WHERE Product_ID = 109;

1 row deleted.

SQL> DELETE FROM Product
  2  WHERE Product_ID = 110;

1 row deleted.
```

9.Category-wise Product Reports

```
SQL> SELECT
2      c.Category_ID,
3      c.Category_Name,
4      COUNT(p.Product_ID) AS Total_Products
5 FROM Category c
6 LEFT JOIN Product p
7 ON c.Category_ID = p.Category_ID
8 GROUP BY c.Category_ID, c.Category_Name
9 ORDER BY c.Category_ID;
```

CATEGORY_ID	CATEGORY_NAME	TOTAL_PRODUCTS
1	Face Care	1
2	Body Care	1
3	Hair Care	1
4	Bath and Shower	1
5	Lip Care	1
6	Skin Care	1
7	Face Masks	1
8	Moisturizers	1
9	Cleansers	0
10	Sunscreen	0

10 rows selected.

10. Product and Category Report

```
SQL> SELECT
2      p.Product_ID,
3      p.Product_Name,
4      c.Category_Name,
5      p.Price
6 FROM Product p
7 JOIN Category c
8 ON p.Category_ID = c.Category_ID
9 ORDER BY c.Category_Name;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_NAME	PRICE
101	Natural Bath Soap	Bath and Shower	199
102	Aloe Vera Body Lotion	Body Care	399
101	Neon Face Mash	Face Care	379
107	Turmeric Face Mask	Face Masks	299
103	Herbal Hair Oil	Hair Care	349
105	Shea Butter Lip Balm	Lip Care	149
108	Coconut Moisturizer	Moisturizers	449


```
-----  
PRODUCT_NAME  
-----  
CATEGORY_NAME                                PRICE  
-----  
106  
Vitamin C Face Serum  
Skin Care                                649  
  
8 rows selected.  
SQL> |
```

11. Expected Category-wise Report

Category	Total Products	Total Stock
Bath and Shower	1	60
Body Care	1	40
Cleansers	1	30
Face Care	1	50
Face Masks	1	25
Hair Care	1	30
Lip Care	1	45
Moisturizers	1	40
Skin Care	1	35
Sunscreen	1	20

The report displays the number of products and total available stock under each category in the Vilvah product and category management system.

12. Conclusion

The Vilvah Product and Category Management System successfully represents the relationships between customers, suppliers, categories, products, orders, order details, payments, inventory and admins.

Primary key and foreign key constraints help maintain data integrity and establish relationships between the tables.

INSERT, UPDATE and DELETE operations support effective product and category management. SQL reports provide useful information about products and stock, while PL/SQL programs can demonstrate procedural database programming for category-wise report generation and product price updates.